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Prevalence of antibiotic resistance gene in different wastewater treatment systems and effluent-irrigated soils through metagenomic analysis

Hua Fang[1,6], Miao Pu[2,4,6], Ailier jiang[1], Fuer haiti[1], Yubei Liu[1], Nuerla Ailijiang[2,3]✉, Anwar Mamat[5] & Xifeng Tu[2,3]

Wastewater treatment systems (WWTS) are considered to be the main source of antibiotic resistance genes (ARGs) spreading into the environment. In this study, samples were collected from WWTS influent, biological treatment tank effluent, and recycled water treatment plant (RTP) effluent during summer and winter, followed by metagenomic sequencing. The study investigated the differences in antibiotic resistance gene transfer between two typical wastewater treatment plants (WWTPs) processes and the impact of recycled water irrigation on ARG dissemination in soil. The WWTS (HD and MD) adopting two combined processes of "Adsorption-Biodegradation Process(AB)+ Anaerobic-Anoxic-Oxic Process(AAO)" and "AAO + Membrane Bioreactor(MBR)" as the research objects for the first time.The primary ARGs types identified were multidrug resistance, tetracycline, macrolide, and aminoglycoside resistance genes. The top three ARGs subtypes by relative abundance in the influent, biological treatment tank effluent, and total effluent were *msrE*, *mphE*, and *ANT(6)-Ia*, respectively. Seasonal variations did not significantly influence the distribution of ARGs in the two WWTSs. The AAO and AB processes increase the relative abundance and diversity of ARGs, while ARGs relative abundance decreases after RTP treatment but may proliferate new ARGs subtypes. Additionally, the efficiency of reducing the relative abundance of ARGs in summer is higher than that in winter. The two WWTSs were able to efficiently remove *msrE* and *mphE*. The abundance and diversity of ARGs and microorganisms were maximum in soil samples from the RTP. The microbial genera significantly related to ARGs may become its potential host, such as *Rhodanobacter* had significant correlations with *ropB2*, *carA*, and *oleB*. These results provide new insights into the control of ARGs contamination and focus on the risks associated with irrigated wastewater.

Keywords Antibiotic resistance gene, Irrigated soil, Reclaimed water treatment plant, Wastewater treatment system

The widespread use of antibiotics for human, animal and agricultural purposes has led to the emergence of antibiotic resistance genes (ARGs) and multidrug resistant microbial communities in water and soil environments[1]. The persistent and worsening problem of antibiotic resistance (AR), which poses a threat to human health and a burden on economic conditions worldwide, has evolved into one of the greatest public health problems of the 21 st century[2,3]. ARGs are not only generated from the human clinical medical environment, but also during the interaction of animals, plants, soils and water bodies[4]. ARGs proliferate by horizontal gene

[1]Xinjiang Communications Investment Construction Management Co., Ltd, Urumqi, China. [2]Key Laboratory of Oasis Ecology of Education Ministry, College of Ecology and Environment, Xinjiang University, Urumqi 830017, P.R. China. [3]Xinjiang Jinghe Observation and Research Station of Temperate Desert Ecosystem, Ministry of Education, Urumqi 830017, P.R. China. [4]Xinjiang Institute of Nonferrous Metals Co, Urumqi 830022, P.R. China. [5]School of Chemical Engineering and Technology, Xinjiang University, Urumqi 830017, PR China. [6]Hua Fang and Miao Pu authors contributed equally. ✉email: aljnel@xju.edu.cn

transfer (HGT) from antibiotic resistant bacteria (ARBs) through mobile genetic elements (MGEs) including plasmids, transposons, and integrins under selective pressure from contaminants such as antibiotics[5].

Wastewater treatment system (WWTS) collect wastewater from hospitals, pharmaceutical industries, and residential households, hence it is considered a hotspot for the spread of AR and a direct source of spreading ARGs and ARBs into the environment[6]. Although WWTSs are primarily employed to remove conventional contaminants, they also play a role in the removal of ARGs, with different WWTSs having variable ARG removal rates due to various parameters[7,8]. In WWTS, the high biomass density, abundant nutrients, adequate oxygen, and residual antibiotics create optimal environment for ARGs propagation via HGT and ARB proliferation. Biological treatment can lead to substantial enrichment of ARGs, ARBs and MGEs, thereby altering sludge microbial communities due to selection of antibiotic residues[9]. Studies have shown that biological treatment processes (e.g., anaerobic digestion) can increase the abundance of ARGs, and significant levels of ARGs can accumulate in sludge[10]. However, research findings on the impact of WWTS on the prevalence of specific ARGs are sometimes contradictory. While some results indicate that WWTS reduce the abundance and diversity of ARGs, thereby lowering the risk of AR transmission, other studies suggest that WWTS serve as a non-negligible source of ARB, contributing to the spread of ARGs[11,12]. Monitoring and controlling the release of antibiotics and ARGs, and ARBs in WWTSs is critical, and to address this issue, it is generally prioritized to determine the occurrence and fate of antibiotics, ARGs and ARBs in WWTSs[13].Furthermore, the reuse of reclaimed water has been identified as one of the primary pathways for the propagation of ARGs[14]. The reuse of treated wastewater to irrigate soil can result in the continued addition of antibiotics, ARGs and ARBs to the environment, and these contaminants can persist in agroecosystems[15]. Parameters related to wastewater and irrigated soil, such as soil texture, pH, organic matter, and nutrient content, are all linked to changes in soil microbial communities and ARGs, implying that these soil characteristics may influence the abundance of ARGs[16]. WWTS can not only develop and sustain ARGs under the selective pressure exerted by antibiotics and heavy metals but also facilitate their transmission to soil and crops through HGT[17].

There are two obvious deficiencies in existing studies: first, there is a lack of systematic comparison on the differential performances of WWTS with different combined processes (such as the combined processes of adsorption-biodegradation process(AB) and anaerobic-anoxic-Oxic process(AAO)) in terms of ARGs removal efficiency, abundance, and subtype evolution, making it difficult to explain the contradictions in research conclusions; second, regarding the impact of reclaimed water irrigation on soil ARGs, the baseline control role of non-irrigated soil is often ignored, and the specific impact of effluent on soil ARGs composition is not analyzed in combination with specific process differences, resulting in the disconnection of the research chain of "process - effluent - soil". Therefore, it is necessary to systematically compare the ARGs response characteristics of different combined process systems and clarify the actual impact of effluent from different processes on soil ARGs in combination with baseline control. The innovation of this study lies in that it takes the WWTS (HD and MD) adopting two combined processes of "AB + AAO" and "AAO + Membrane Bioreactor(MBR)" as the research objects for the first time, systematically compares their differences in ARGs removal efficiency, abundance, diversity, and subtype changes, and at the same time, takes non-irrigated soil (NS5) as the control, and combines the process characteristics of the two systems to deeply analyze the specific impact of their effluent irrigation on soil ARGs composition and potential host microorganisms, thus filling the research gap in the chain of "process differences - effluent ARGs characteristics - soil responses".

This study focused on two WWTSs in a semi-arid region, which adopt the "AB + AAO" and "AAO + MBR" processes respectively. Using metagenomic sequencing, it analyzed the distribution characteristics of ARGs annotated in the Comprehensive Antibiotic Resistance Database (CARD) and the composition of microbial communities. The specific objectives were as follows: (I) To clarify the differences in abundance and diversity of ARGs in various treatment stages of the two systems and in reclaimed water-irrigated soils; (II) Based on samples collected in December 2020 (winter) and June 2021 (summer), to evaluate the effects of biological treatment and reclaimed water treatment plants (RTPs) on the removal efficiency and subtype evolution of ARGs in different seasons; (III) To analyze the correlations between ARGs and bacterial communities as well as antibiotic concentrations, and identify potential host microorganisms of ARGs. In the study, winter and summer water samples from both systems were collected from the same sampling points (influent, effluent from biological treatment tanks, and final effluent from RTPs), with only differences in sampling time (e.g., WHD and SHD of WWTS-HD, WMD and SMD of WWTS-MD). This ensured the comparability of data between seasons and provided a reliable basis for analyzing the impacts of processes and seasons on ARGs.

Materials and methods

Sampling

The target WWTSs (namely "HD" and "MD", 'HD' and 'MD' are abbreviations for the region where the system is located, aiming to simplify the expression and avoid involving specific geographical names) are in Urumqi, Xinjiang, northwest China, where winter lasts from November to April. Each wastewater treatment system consists of a WWTP and a RTP. WWTP-HD adopts the AB process: activated sludge first adsorbs organic pollutants, and then microorganisms decompose these pollutants. WWTP-MD uses the AAO process, which treats wastewater through sequential anaerobic, anoxic, and oxic zones to remove organic pollutants. For reclaimed water treatment, RTP-HD employs the AAO process to further purify the effluent from WWTP-HD, while RTP-MD combines AAO with a MBR. The MBR separates sludge and water through membranes, retains microorganisms, improves efficiency.During the experiment, we took three duplicate samples from each sample point and then condensed them into one sample for testing.

Wastewater samples were collected from the two WWTSs in December 2020 (winter) and June 2021 (summer). Samples were collected from the influent, biological treatment tank effluent, and RTPs final effluent. The detailed processes of the WWTSs and the sampling points are shown in Fig. 1. Wastewater samples were

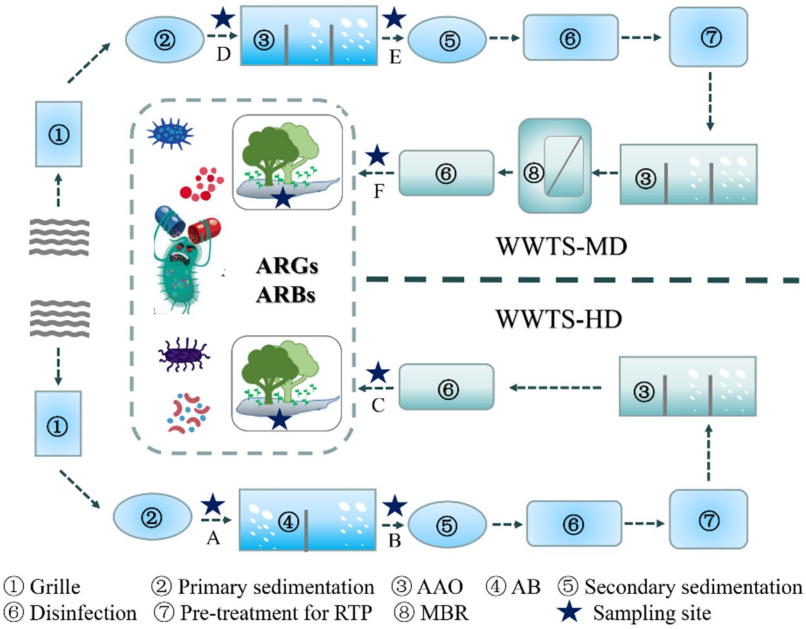

Fig. 1. Diagram of two wastewater treatment systems with different reclaimed water treatment processes. (AAO refers to the Anaerobic - Anoxic - Oxic process, AB is the Adsorption - Biooxidation process, and MBR stands for Membrane Bioreactor. WWTS - MD represents the wastewater treatment system adopting the AAO + MBR process; WWTS - HD represents the wastewater treatment system adopting processes such as AB + AAO. The star marks are the sampling sites in this study.).

collected in sterile amber bottles that were rinsed with water prior to sampling. The microbial biomass were concentrated by vacuum extraction and filtration within 4 h after sampling, and the dried filter membrane with enriched bacteria was folded and stored in a 5 mL sterile EP tube.

The effluent from each RTP is used for green irrigation—defined here as irrigation of non-agricultural green spaces, including urban parks, lawns, and roadside vegetation—which is applied from approximately April to November each year (i.e., not in winter). Considering key sampling factors—such as consistent irrigation history, homogeneous soil type, and proximity to the reclaimed water outlet (to ensure uniform exposure to effluent from RTP-HD)—the reclaimed water irrigation area associated with WWTS-HD was selected as the research object. Soil samples were collected in June 2021 (Table S1), encompassing three distinct categories based on their location and irrigation status: (1) samples from within reclaimed water plant areas (RS1), referring to soils within the premises of the reclaimed water treatment plant; (2) samples from effluent-irrigation areas (IS2, IS3, and IS4), representing soils in regions regularly irrigated with reclaimed water effluent; and (3) samples from non-irrigation areas (NS5), denoting soils in areas with no history of reclaimed water irrigation (serving as a reference).Soil samples were collected in June 2021 (Table S1), and the wastewater and soil samples were collected with the permission of the local WWTP operators. A total of 15 soil samples were randomly collected from the upper 20 cm of the soil profile at five sampling points using a stainless-steel soil sampler, with three samples from each point forming a mixed sample. After being transported in iceboxes to the laboratory, samples were separated into a sterile EP tubes after being screened by 2 mm. All samples were stored at −80℃ until metagenomic analysis. WHD and WMD denote winter WWTS-HD and WWTS-MD wastewater samples, respectively. SHD and SMD denote summer WWTS-HD and WWTS-MD wastewater samples, respectively. RS1, IS2-IS4 and NS5 denote soil samples.

Method for detecting antibiotics

The method for detecting antibiotic concentrations in this paper is as follows: All target antibiotics were separated using ACQUITY UPLC BEH C18 columns (2.1 mm × 100 mm, 1.7 μm particle size) with different gradients, and detected by electrospray ionization (ESI)-tandem mass spectrometry (MS-MS). Antibiotic concentrations were quantified using the external standard method, and 100 ng/L of standard was added to each sample to calculate the recovery rate of the target antibiotics. Samples were processed according to the above method to ensure the reliability of the experimental data. The recoveries of blank water samples and standard water samples ranged from 61.30% to 152.4% with RSD < 15% (Table S3). The matrix effect values of the target antibiotics were 88–119%. The minimum detection limit was measured at a signal-to-noise ratio of 3, and the minimum quantification limit was measured at a signal-to-noise ratio of 10. Detailed information is provided in Table S3. Therefore, this method has good reproducibility, and its precision and accuracy meet the requirements for the analysis of antibiotics in samples.

DNA extraction and ARGs detection

The genomic DNA was extracted from samples using PowerSoil/PowerWater DNA Isolation Kit (MOBIO, USA, PowerSoil® DNA Isolation kit) according to the manufacturer's instructions. The quality and concentration of the DNA were detected using Nanodrop spectrophotometer (Nanodrop ND-2000). The extracted DNA samples were immediately stored at − 20℃ and then until metagenomic analysis.

The genomic DNA was fragmented 300 bp using ultrasonic breaker Covaris M220 (Gene Company, China), and paired-end library was prepared using NEXTFLEX Rapid DNA-Seq Kit (Bioo Scientific, Austin, TX, USA). All samples were sent to Majorbio Bio-Pharm Technology Co., Ltd (Beijing, China) for Illumina NovaSeq sequencing. All water samples were filtered before extraction. The specific filtration parameters are as follows: an aluminum membrane filter with a pore size of 0.45 μm was used; for the effluent samples from the reclaimed water plant, which had relatively clear water quality, the filtration volume was approximately 6–7 L; for other water samples with higher turbidity, the filtration volume was about 1.5 L; and the volume of all water samples submitted for testing was uniformly 500 mL.

Bioinformatics analysis

The raw reads contained low quality sequences and data filtration was performed to guarantee the quality of downstream analysis. FASTQ files were trimmed for low quality bases and adapters using Trimmomatic (High-throughput sequencing analysis of the bacterial 16 S rRNA gene was performed using the Illumina Novaseq 6000 SP PE250 platform. The V3–V4 regions of the bacterial 16 S rRNA gene were amplified by PCR using primers 341 F (CCTACGGGNGGCWGCAG) and 805 R (GACTACHVGGGTATCTAATCC). Clean Reads were obtained by quality filtering of sequenced Raw Reads using FASTQ (V0.14.1), USEARCH (V10.0.240) was used for splicing and filtering the Clean Reads, and Effective Reads were obtained by DADA2 denoising. All Effective Reads were clustered into operational taxonomic units (OTUs) by UCLUST with a similarity threshold value of 97%, and 16 S Silva V132 database was used for annotation, and 16 S Silva V132 database was used for annotation.) to remove reads shorter than 100 bp. The metagenomic sequencing data were assembled using MEGAHIT[18](Version 1.1.2), and contings ≥ 300 bp were selected for final assembly. The assembled reads were used for further gene prediction and annotation. MetaGeneMark[19](Version 3.26) was used to identify the coding regions in the genome using default settings. The motifs ≥ 100 bp were selected and translated into amino acid sequences. Nonredundant gene catalog was constructed using CD-HIT[20] (Version 4.6.6) with 95% identity and 90% coverage. SOAPaligner[21] was used to locate all quality controls into a non-redundant gene catalog with 95% identity and to calculate abundance information of gene in the corresponding sample. The taxonomic classification was obtained NR database by Diamond (Version 0.9.24) with a maximum e-value threshold of 1e-03. The amino acid sequence of the nonredundant gene catalog was compared with CARD (the Comprehensive Antibiotic Research Database) sing RGI (Version 4.2.2) and obtained the annotation information. The portion of types or subtypes of ARGs sequences in the 'total metagenome sequences' were defined as the 'relative abundance' ('p.p.m.', one read in one million reads). In this study, the relative abundance of ARGs (in ppm) was normalized by "the number of sequences annotated as ARGs per million sequencing sequences" to eliminate the impact of differences in sequencing depth among different samples on the results. The removal rate of ARGs is calculated using Eq. (1):

$$\text{Removal efficiency}\ (\%) = \frac{\text{Abundance}_{\text{Inflluent}} - \text{Abundance}_{\text{Effluent}}}{\text{Abundance}_{\text{Inflluent}}} \tag{1}$$

Results and discussion

Occurrence of ARGs in WWTSs

Metagenomic analysis revealed that the relative abundance of total ARGs showed divergent trends following biological treatment of primary wastewater (Fig. 2A). In WWTS-HD, there was a consistent increase in total ARGs abundance, a phenomenon closely linked to the unique microenvironmental conditions within its biological treatment tanks—characterized by abundant nutrients, high microbial biomass, and selective pressures from residual antibiotics and heavy metals, which create a hotspot for HGT among microorganisms and facilitate the dissemination of ARGs and proliferation of ARB[22,23]. However, WWTS-MD (WMD) exhibited the opposite trend, with a reduction in total ARGs abundance post-biological treatment. This discrepancy may be attributed to system-specific factors, such as differences in microbial community composition, operational parameters (e.g., aeration intensity or sludge retention time), or influent water quality in WWTS-MD, which could mitigate ARGs proliferation during biological treatment.

In contrast, subsequent treatment by RTPs led to a notable reduction in ARGs abundance across both systems, highlighting their critical role in mitigating ARGs loads in effluents. However, a more nuanced pattern emerged when examining ARG diversity alongside abundance: RTP treatment resulted in opposing trends for these two metrics (Fig. 2A&B). This divergence suggests that while RTPs effectively lower overall ARGs levels, the selective pressure exerted by their treatment processes (e.g., membrane filtration in MBR or nutrient adjustment in AAO) may drive the emergence of novel ARG variants—potentially adapting to survive these specific conditions. Intriguingly, during summer, biological treatment stages showed a decrease in ARG diversity, which may reflect seasonal shifts in microbial community structure (e.g., altered metabolic activity under higher temperatures) that restrict the range of ARGs capable of persisting or spreading.

Notably, neither the AB + AAO system (WWTS-HD) nor the AAO + MBR system (WWTS-MD) achieved complete elimination of ARGs relative abundance, with residual ARGs posing substantial risks for subsequent water reuse and irrigation—exacerbating concerns in regions already facing water scarcity[23,24]. While seasonal variations did not exert a statistically significant impact on the wastewater resistome (Table S2, ANOVA, $p >$ 0.05), winter samples consistently exhibited slightly higher ARG diversity and relative abundance compared to

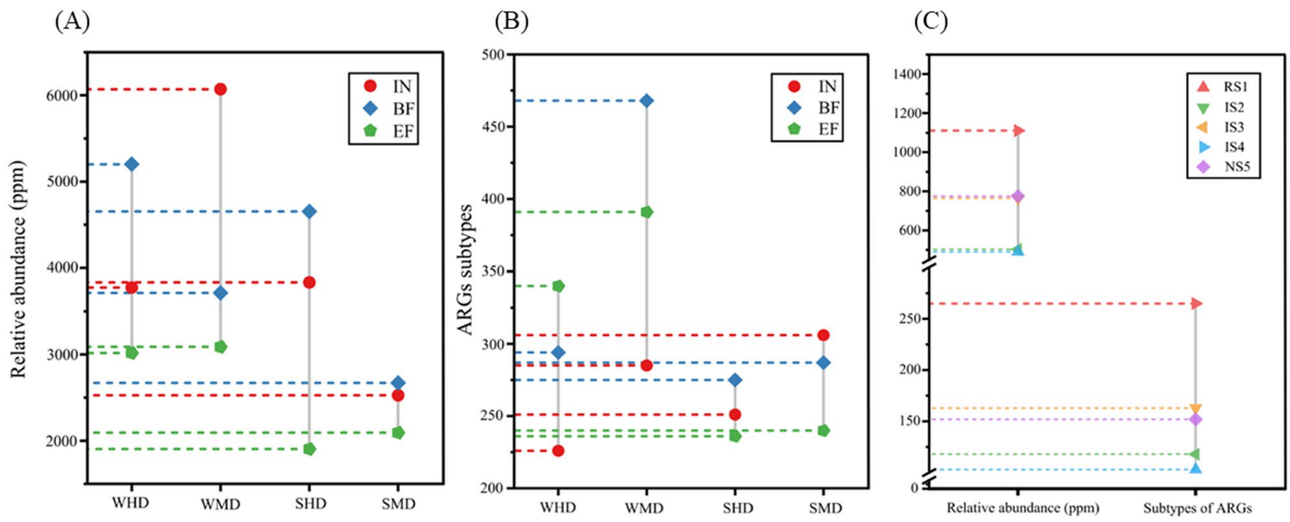

Fig. 2. Relative abundance (**A**) and subtypes (**B**) of antibiotic resistance genes (ARGs) in different wastewater types, and relative abundance of ARGs in different soil types (**C**)". (WHD: wastewater samples from the WWTS-HD system collected in Winter; WMD: wastewater samples from the WWTS-MD system collected in Winter; SHD: wastewater samples from the WWTS-HD system collected in Summer; SMD: wastewater samples from the WWTS-MD system collected in Summer; IN: influent; BF: biological effluent; EF: effluent; RS: soils in the Reclaimed Water Treatment Plant (RTP); IS: irrigated soils; NS: non-irrigated soils.).

summer (Fig. 2A&B), with elevated antibiotic resistance determinants detected in winter final effluents. This subtle seasonal pattern may relate to reduced microbial activity in colder temperatures, which could slow ARG degradation, or increased antibiotic usage in winter (e.g., for treating seasonal illnesses) that amplifies selective pressures.

Metagenomic sequencing identified 31 distinct resistance types across all water samples, with multidrug, tetracycline, macrolide, and aminoglycoside resistance types dominating (Fig. 3)—a pattern consistent with findings from global wastewater studies[25]. This prevalence likely stems from the widespread use of these antibiotic classes in human and veterinary medicine, leading to their frequent introduction into wastewater systems. Multiple interconnected factors may further shape this resistance profile, including antibiotic-pathogen co-selection, cross-resistance mechanisms, geographic-specific antibiotic usage patterns, and local demographic factors (e.g., population density, sanitation infrastructure, and vaccination rates)[26].

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Analysis of the top 15 ARG subtypes by relative abundance (Fig. 4) revealed substantial variability in subtype richness across treatment stages: WWTS-HD showed 226–251, 275–294, and 236–340 subtypes in influent, biological treatment effluent, and final effluent, respectively, while WWTS-MD exhibited 285–306, 287–468, and 240–391 subtypes in the same stages. Dominant subtypes included *msrE*, mphE, *ANT(6)-Ia*, *tet(E)*, *adeF*, and *tet(39)*—aligning with the high abundance of their corresponding resistance types (multidrug, tetracycline, aminoglycoside). Notably, *msrE* and *mphE* abundances often increased after AAO or AB treatment but declined sharply post-RTP, suggesting RTP processes (e.g., membrane filtration in MBR or enhanced biological activity) specifically target these subtypes. The strong association between *msrE* and diverse microbial communities, coupled with evidence that wastewater discharge enriches sediment with *msrE*[27]. highlights its potential for environmental dissemination. Conversely, *tetQ*, *cmeB*, and *ANT(6)-Ia* showed marked abundance reductions across treatments, indicating greater susceptibility to the applied processes.

Process-specific effects on ARG subtypes were also observed: in WWTS-HD, AB treatment significantly increased *chrB*, *BEL-1*, *tet(E)*, and *Erm(38)* abundances, while RTP treatment elevated adeF and *IMP-22* relative to influent. In WWTS-MD, AAO + MBR treatment boosted *tet(Y)*, *OXA-3*, and *chrB* levels—collectively indicating that treatment processes can inadvertently promote HGT, increase ARG abundance, and drive the emergence of new genotypes.

Co-occurrence and uniqueness of ARG subtypes across *samples* (Fig.S1) further illuminated treatment limitations: 33–72 subtypes persisted across all WWTS-HD stages, and 80–92 across WWTS-MD stages, indicating incomplete removal. Meanwhile, 64–153 subtype differences between consecutive stages reflected dramatic shifts in ARG diversity, and unique subtypes in each stage suggested biological and RTP treatments may facilitate ARG proliferation via HGT. These patterns collectively emphasize the need for optimized treatment strategies to better control ARG persistence and spread in reclaimed water systems.

Removal of ARGs in WWTSs

Figure 3 delineates seasonal variations in the dominant ARG types across treatment stages of the WWTSs during winter and summer, revealing a generally uniform distribution of resistance types—with notable exceptions where specific categories dominate. Among all ARG types, multidrug resistance genes consistently constitute the largest fraction, and their persistence underscores a critical limitation of current treatment processes: neither

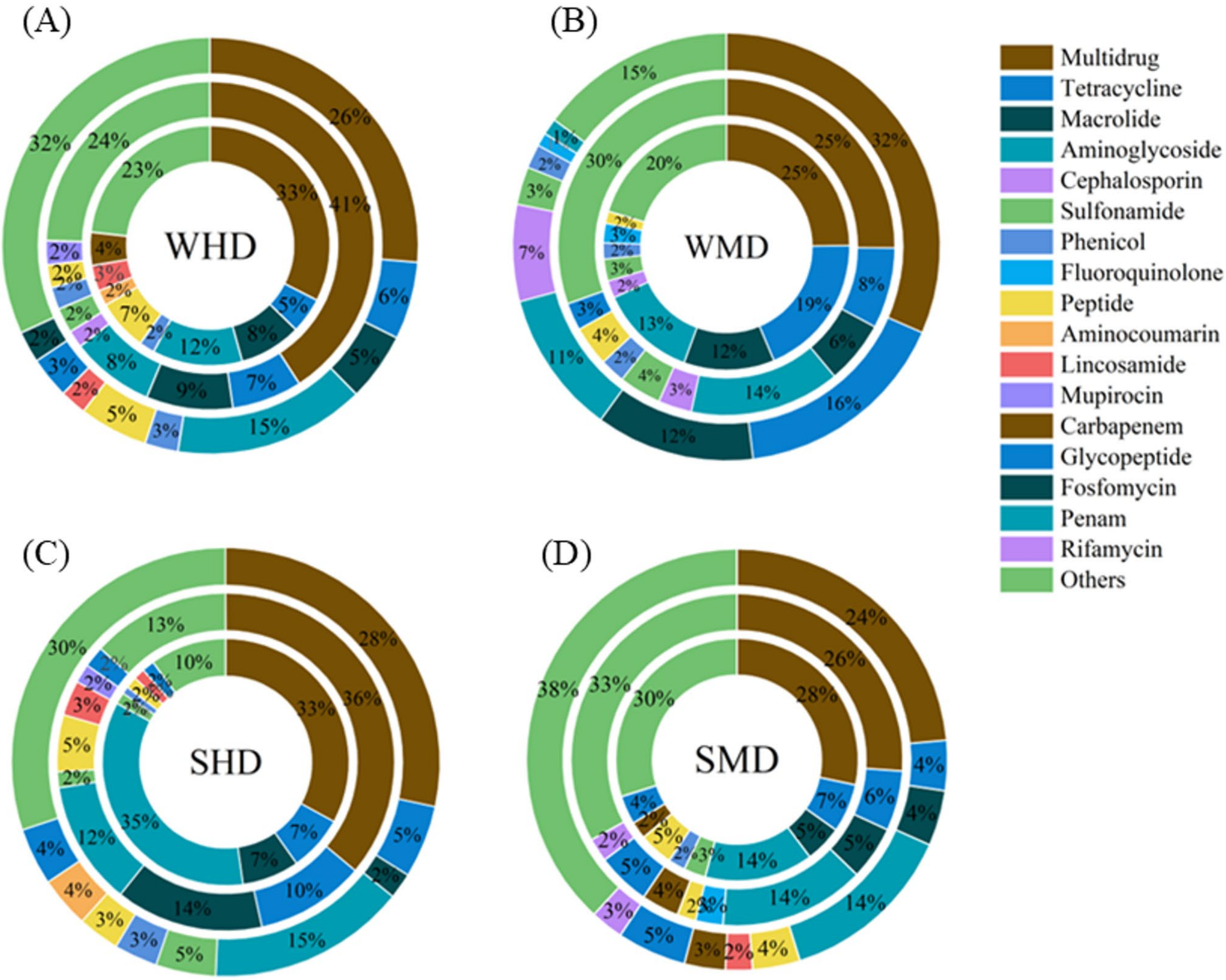


Fig. 3. Distribution of ARGs in different wastewater treatment processes ((**A**) WHD represents wastewater from the WWTS - HD system collected in winter; (**B**) WMD represents wastewater from the WWTS - MD system collected in winter; (**C**) SHD represents wastewater from the WWTS - HD system collected in summer; (**D**) SMD represents wastewater from the WWTS - MD system collected in summer. In each subfigure, the inner to outer circles correspond to influent, biological effluent, and effluent respectively. The legend on the right indicates different ARG categories).

the AB nor AAO biological treatments effectively reduce their abundance, and in some cases, these processes even enhance it. This phenomenon likely stems from the broad adaptability of multidrug resistance mechanisms, which enable survival under diverse selective pressures (e.g., co-occurrence of multiple antibiotics or heavy metals) within biological reactors. Furthermore, the RTP stages paradoxically contribute to the dissemination of multidrug ARGs, s uggesting that while RTPs mitigate overall ARG loads, they may exert unique selective pressures that favor the persistence of these versatile resistance determinants.

Deeper scrutiny of process-specific effects reveals nuanced patterns in ARG removal. In WWTS-HD, the AB process reduced aminoglycoside ARGs by 3%–23%, whereas in WWTS-MD, the AAO process achieved modest reductions in tetracycline (10%) and macrolide (6%) ARGs—indicating that certain resistance types are more susceptible to specific biological treatments. A notable success across both systems is the effective removal of *msrE* and *mphE*, with their relative abundances decreasing by 1–2 orders of magnitude post-treatment. This suggests that these subtypes (associated with macrolide resistance) are particularly vulnerable to the combined effects of biological degradation and subsequent RTP processing. Conversely, the AB process exhibited a dual role: while it reduced abundances of *ANT(6)-Ia*, *cmeB*, *tet36*, and *tetQ*, it simultaneously proliferated ARGs such as *adeF*, *Erm(38)*, *tet(E)*, *BEL-1*, and *chrB*. This dichotomy highlights the selective nature of biological treatment, where microbial community shifts or HGT may favor the expansion of certain resistance determinants even as others are suppressed. Similarly, the AAO process reduced *tetQ*, *tet(39)*, *adeF*, and *blaR1*, but RTP stages further lowered abundances of *msrE*, *mphE*, *BEL-1*, and *chrB*—reinforcing the RTP's role as a critical secondary barrier.

Overall, the AB and AAO biological processes demonstrate limited efficacy in ARG removal, with the latter emerging as a significant driver of ARG enrichment. This aligns with broader observations that biological treatment units, characterized by high biomass density, abundant nutrients, and persistent selective pressures

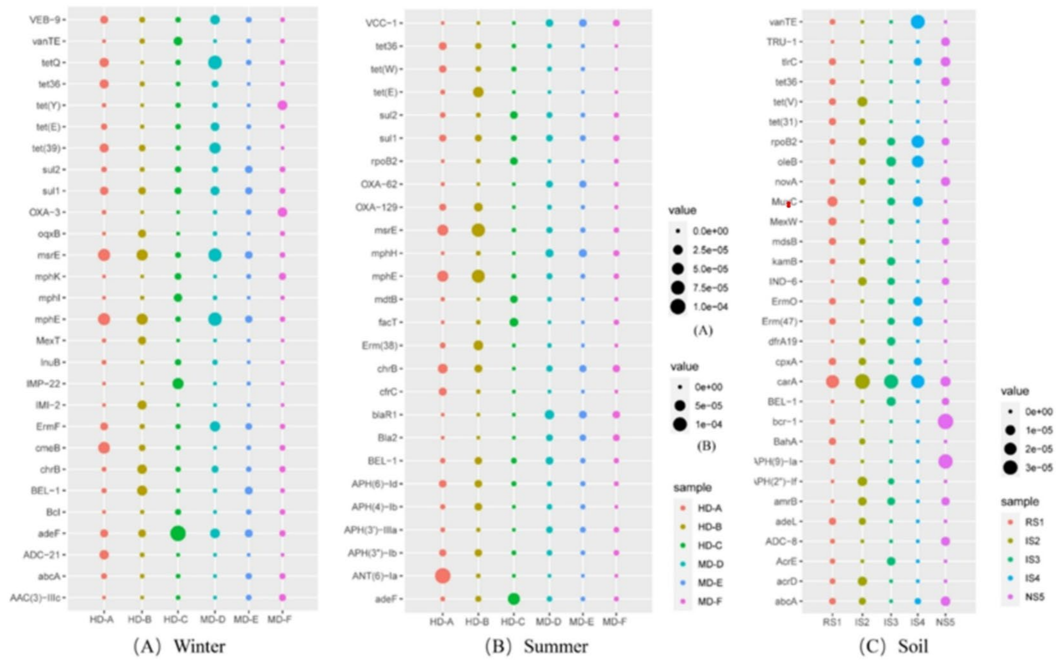


Fig. 4. Relative abundance of top 25 ARGs subtypes in samples(WHD represents wastewater samples from the WWTS - HD system collected in winter; WMD represents wastewater samples from the WWTS - MD system collected in winter; SHD represents wastewater samples from the WWTS - HD system collected in summer; SMD represents wastewater samples from the WWTS - MD system collected in summer. RS1 refers to soils in the Reclaimed Water Treatment Plant (RTP); IS2, IS3, IS4 refer to irrigated soils; NS5 refers to non - irrigated soils.).

from residual antibiotics or heavy metals, create optimal conditions for the proliferation of ARBs and the spread of ARGs via mechanisms like transduction[28]. In contrast, RTP stages achieved varying degrees of ARG reduction: RTP-HD lowered tetracycline and macrolide ARGs by 1%–5% and 3%–12%, respectively, while RTP-MD removed 1%–3% of aminoglycoside ARGs. These reductions are hypothesized to stem primarily from disinfection steps: RTP-A employs ultraviolet (UV) disinfection, which penetrates cell membranes to target DNA/RNA, disrupting nucleic acid structures and inactivating ARGs[29,30]. RTP-B uses ozone disinfection, where O_3 reacts with pollutant functional groups or free radicals to degrade ARGs[30,31]. The efficacy of these methods is context-dependent: UV performance varies with ARG type, exposure duration, UV dose, and microbial community composition[23], while ozone's effectiveness hinges on reaction kinetics with specific resistance determinants.

Consistent with findings from other studies, biological treatment processes tend to proliferate ARGs (Fig. 5)[9,22]. The high biomass density, abundant nutrients, and selection pressure from antibiotics, heavy metals, or other contaminants within biological treatment units create an optimal environment for the proliferation of ARBs and the transfer of ARGs[22,23]. In contrast, the relative abundance of ARGs decreased after RTP treatment (Fig. 5A). Disinfection is expected to play a major role in ARGs removal. The biological removal efficiency of total ARGs abundance ranged from − 37.76 to 38.83% and the total removal efficiency ranged from 17.17 to 50.35% (Fig. 5B). Overall, the AAO removed ARGs better than the AB process, probably due to the variable environmental conditions in the AAO, which were more favorable to the removal of ARGs by a variety of microorganisms. Even though there was a negative removal of ARGs, the removal efficiency increased in summer compared to winter.

Occurrence of ARGs in soils samples

In the design of this study, the non-irrigated soil group (NS5) was selected as a relative background control to comparatively evaluate the impact of wastewater irrigation on soil ARGs. The research focused on the differences in ARG removal efficiency between the two wastewater treatment systems. Among them, the HD system, due to its higher pollutant concentration in effluent and relatively poor removal rate, its irrigation area was prioritized as the soil research object. Through comparison with NS5, the potential impact of irrigation on soil ARGs was analyzed. For soil samples, the total relative abundance of ARGs was 491–1111 ppm, and a total of 103–265 ARGs subtypes were detected (Fig. 2C). The samples within the RTP (RS1) showed the highest abundance and diversity of ARGs. Compared with other samples, the irrigation frequency of reclaimed water at this point was higher. It is speculated that the frequent irrigation promotes the proliferation of ARGs in the soil to a certain extent or that the residual ARGs in the reclaimed water enter the soil and undergo HGT. The abundance of ARGs in non-irrigated soils is higher than that in some irrigated soils. The reasons may include that the residual antibiotics in the solid waste accumulated around the site generate selective pressure to promote the diffusion

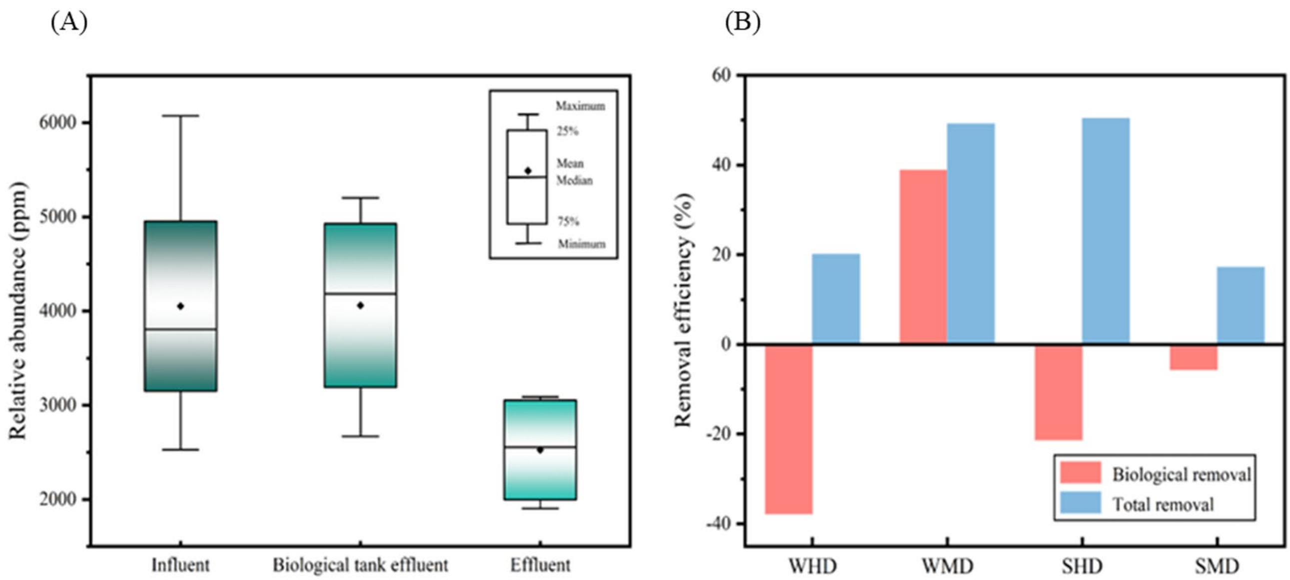


Fig. 5. (**A**) Changes of ARGs distribution; (**B**) Removal efficiency of ARGs(WHD represents wastewater samples from the WWTS - HD system collected in winter; WMD represents wastewater samples from the WWTS - MD system collected in winter; SHD represents wastewater samples from the WWTS - HD system collected in summer; SMD represents wastewater samples from the WWTS - MD system collected in summer.).

of ARGs[24], and may also be due to the existence of natural ARGs in the soil environment[32]. Irrigation with reclaimed water did not have a significant impact antibiotic resistance in soils over a short period of time. There were no significant differences in ARGs between the wastewater irrigated and non-irrigated areas in this study (Table S3, ANOVA, $p > 0.05$ The p-value of 0.6424 is far greater than 0.05, indicating that soil type has no significant impact on the abundance of resistance genes), but irrigation may have influenced the composition of ARGs. The NS5, serving as a control, provides a crucial reference for assessing the potential impact of irrigation through its baseline levels of ARGs. Comparative analysis reveals that although irrigation did not significantly alter the overall abundance of ARGs in the soil, there were distinct differences in the proportions of specific ARG subtypes (such as multidrug and tetracycline resistance genes) between irrigated soils and NS5. For instance, the relative abundance of tet(E) in IS2 was 12.3% higher than in NS5. This indicates that wastewater irrigation may alter the compositional structure of soil ARGs by selectively enriching specific resistance gene subtypes. Such control-based analysis further reveals that the impact of irrigation on soil ARGs is not merely a simple increase or decrease in abundance, but a more complex reshaping at the community structure level, offering a more detailed perspective for understanding the association between wastewater irrigation and the evolution of soil resistance genes. The similar composition of ARGs and ARBs indicates that after the reclaimed wastewater enters the soil, the ARBs in the wastewater cannot compete or survive with the local microorganisms, so they cannot promote the HGT[32].

Similar to the resistance types in wastewater, the abundance of multidrug types in soil was the highest, followed by tetracycline, macrolide, and aminoglycoside ARGs. The relative abundance of ARGs subtypes in the top 15 in each soil sample is shown in Fig. 4C. The relative abundance of ARGs in soil samples from RTP was higher than non-irrigated soils. *carA*, *ropB2*, *oleB*, and *bcr-1*, *APH(9)-Ia*, *carA* were the main ARGs subtypes in irrigated soils and non-irrigated soils, respectively. These results suggest that agricultural activities can enrich ARGs. Studies have shown that the abundance of ARGs in the reclaimed water irrigated area is significantly higher than that in the non-irrigated area. Although the abundance of genes related to HGT is not significantly different between the irrigated and non-irrigated areas, the abundance and diversity of ARGs are different, which indicates that the effect of reclaimed water irrigation on soil is obvious[33]. However, in other studies it was found that wastewater irrigation did not affect resistance levels in the soil[34]. Therefore, regardless of whether wastewater irrigation has an impact on the soil environment, the problem of ARGs pollution in the soil environment has widely emerged, requiring attention and control. Even if the use of reclaimed water to irrigate the soil is stopped, the relative abundance of ARGs in the soil does not decrease after a period of time[35].

Bacterial community

At the phylum level, a total of 186 bacterial phylum were detected. It has been shown that the distribution of microorganisms in WWTSs is similar at the phylum level, with the main bacterial phyla including *Proteobacteria*, *Bacteroidetes*, *Actinobacteria*, *Chloroflexi*, and *Firmicutes*[36]. Figure 6 shows the top 20 microbial genera in abundance. *Arcobacter*, *Sphingomonas*, *Acinetobacter*, *Rhodanobacter*, *Methylotenera* and *Pseudomonas* were the dominant genera with higher relative abundance. *Arcobacter* belongs to *Proteobacteria*, and *Arcobacter spp.* associated with high levels of faecal pollution can be used as indicators of pollution in receiving industrial

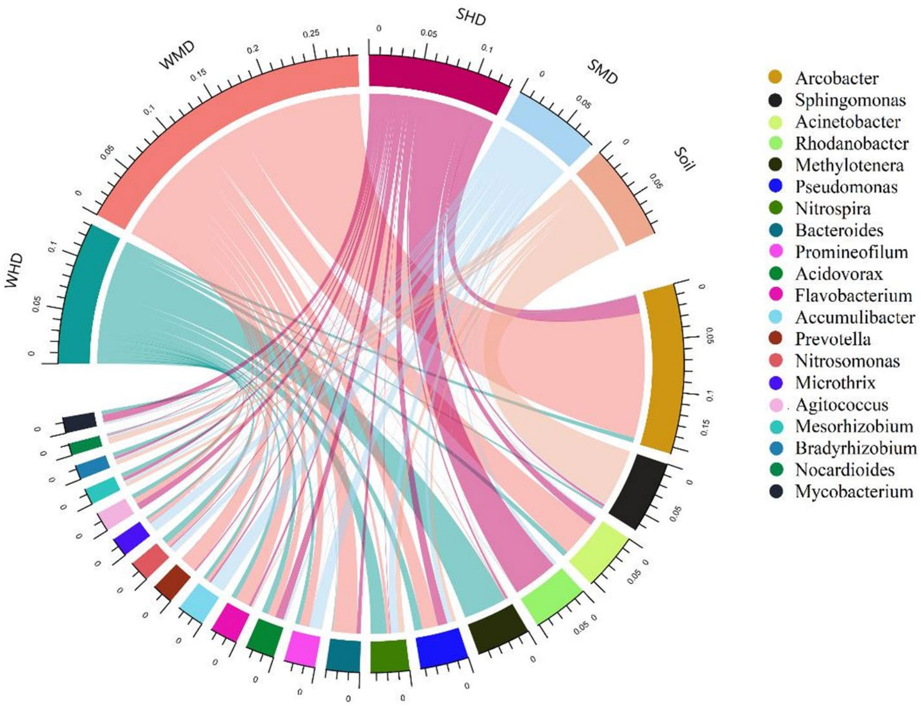
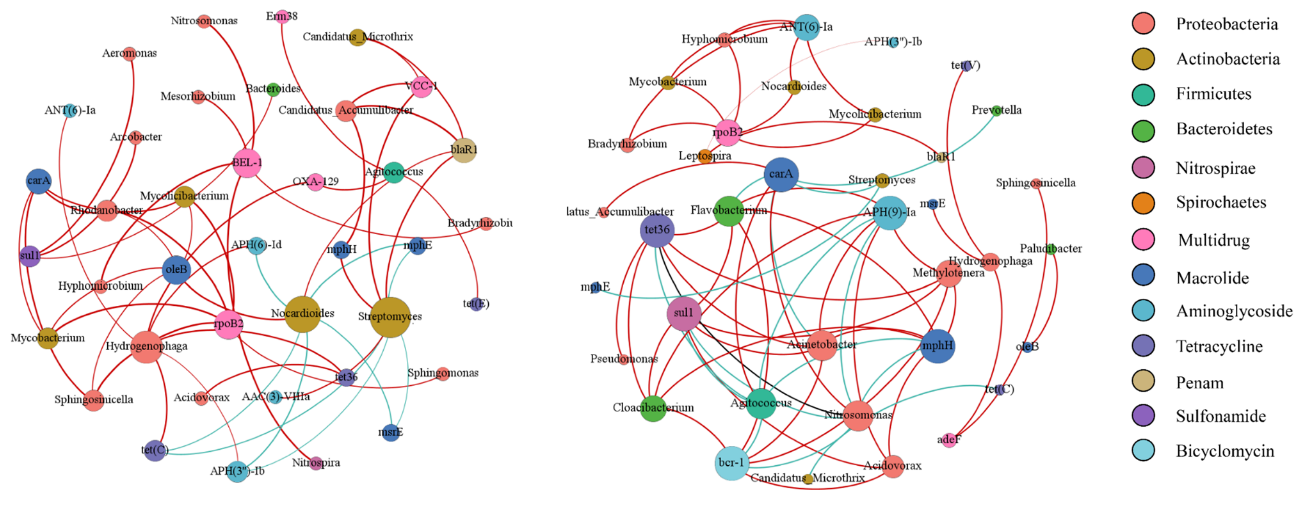


Fig. 6. Composition circle of the top 20 microbial genera in terms of abundance in the sample(WHD represents wastewater samples from the WWTS - HD system collected in winter; WMD represents wastewater samples from the WWTS - MD system collected in winter; SHD represents wastewater samples from the WWTS - HD system collected in summer; SMD represents wastewater samples from the WWTS - MD system collected in summer.).

effluent watersheds and as sanitary indicators for drinking water sources[37]. *Arcobacter* is not only pathogenic to humans and animals leading to disease transmission, but also has the potential to be a host for ARGs and MGEs[38]. *Sphingomonas* was found to be a host for the MGEs *ISSsp4*[39]. A number of bacterial genera associated with nitrogen and phosphorus removal, such as *Pseudomonas*, *Acidovorax*, *Nitrospira*, *Nitrosomonas* and *Sphingomonas*, have been found to host many ARGs, making ARGs widely distributed in WWTSs through HGT[38]. *Acinetobacter* is a human pathogen and has been listed as a high-risk genus for acquiring multidrug resistance genes[25]. Due to the existence of a large number of potential pathogens and ARGs, wastewater needs to be treated to reduce the harm to the environment and humans. The higher abundance of *Acinetobacter*, *Pseudomonas* and *Bacteroides* as potential pathogens in the two WWTSs may became hosts of different ARGs[38]. Nevertheless, there were no significant differences in the microbial communities of the two WWTSs between seasons, suggesting that season variation did not have a significant effect on the distribution of microbial genera (Table S4, ANOVA, $p > 0.05$).

Network analysis

The possible relationships between ARGs subtypes and microbial communities were explored through network analysis to reveal which microbes could be potential hosts of ARGs. Figure 7 shows the network analysis diagram between the microbial community genera and ARGs subtypes obtained by the Pearson algorithm with $p < 0.05$. It can be seen from the network nodes that microbial communities can carry multiple ARGs at the same time, and the same ARGs can choose different microbial communities as hosts (Fig. 7A). For example, *BEL-1* and *oleB* had significant positive correlations with at least five microbial genera; *Rhodanobacter* had significant correlations with *ropB2*, *carA* and *oleB.* Changes in the microbial community will directly affect the changes in ARGs[11]. Changes in bacterial abundance are an important factor affecting the occurrence of ARGs: the higher the bacterial abundance, the higher the associated ARGs abundance[38]. ARGs and coexisting microbial communities showed similar abundance trends in different samples, and this non-random coexistence pattern could provide potential host information for ARGs[40]. The figure shows that the microbial communities that are significantly associated with ARGs may become their hosts. Other functional microorganisms with higher frequency may carry ARGs, such as *Flavobacterium* carrying *tetH*[41]; *Nitrosomonas* carrying *EF-TU*, *penA*, *oqxBgb*, *vanHAc2*, *vanR-F* and *dfr-K*[11,42]; *Dechloromonas* carrying *pep-EC*, *oqx-A*, *oqxBgb*, *vanR-Pt*, and *dfrA3*[43]; *Clostridium* carries *vanR-Pt*, *vgaB* and *tetO*[11]. HGT in native and environmental colonies and selection pressure from heavy metals or antibiotics can lead to the spread of ARBs[11]. Network analysis provides new insights and effective complement on the correlation between ARGs and microbial communities, but the robustness of using network analysis to predict the host of ARGs needs further verification[44,45].



(A) Wastewater samples (B) Soil samples

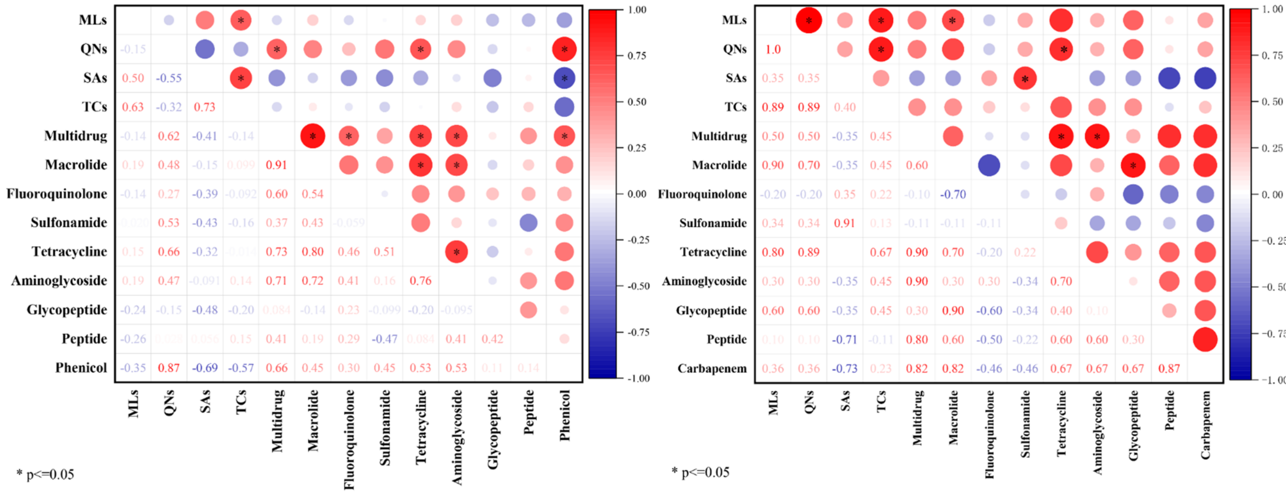
Fig. 7. Network analysis, (**A**) Wastewater samples; (**B**) Soil samples.

Figure 7(B) shows the results of network analysis in soil samples. Gene of the *rpoB2* was positively correlated with five microbial communities: *Bradyrhizobium*, *Nocardioides*, *Mycobacterium*, *Hyphomicrobium*, *Leptospira* and *Mycolicibacterium*, which may become its potential hosts; *APH(9)-Ia*, *bcr-1* and *sul1* were significantly positively correlated with at least three microbial genera. Overall, the relationship between ARGs and microbial communities in wastewater samples is more complex. One reason is that the level of antibiotic resistance in soil is lower than that in wastewater, ARGs only spread among a small number of microbial through HGT. Another reason may be the low abundance of MGEs in soil and the low mobility of soil resistance, which limit the possibility of HGT of ARGs[46,47]. However, under the selective pressure of residual antibiotics in irrigation wastewater, the microbial populations in wastewater and soil are mixed, which will create a favorable environment for the transfer of ARGs between different communities and promote the spread of antibiotic resistance[48]. Especially affecting the abundance, diversity and migratory capacity of ARGs and MGEs in a short period of time[49], the horizontal transfer of ARGs leads to the existence of more potential hosts.

Correlation analysis between antibiotic concentration and ARGs

The release of antibiotics and their metabolites into the environment is an important factor in the evolution and selection of antibiotic resistance in pathogens. The continuous input of ARGs and ARBs from external point and non-point sources, such as WWTSs, is thought to be an important factor driving the emergence of elevated levels of ARGs in places where antibiotics are not present[50]. The method for detecting antibiotics in the article on antibiotic concentration is: All target antibiotics were separated using ACQUITY UPLC BEH C18 (2.1 mm × 100 mm, 1.7 μm particle size) columns with different gradients, and detected using ESI–MS–MS. The column was set at 40 °C at a flow rate of 0.4mL/min, and the injection volume was 5μL. The mobile phase consisted of 5mmol/L ammonium acetate (A) and acetonitrile (B). The gradient was set as follows: 0–0.25 min, 90% A; 0.25–3 min, 10–90% A; 3–4 min, 10% A, 4–5 min, 10–90% A; post time of 5 min. The instrument conditions were as follows: capillary voltage: 4 kV; drying gas temperature: 600 °C; drying gas flow: 3 L/min; and nebulizer pressure: 35psi. Antibiotic concentrations were quantified using the external standard method, and the recovery of the target antibiotics was calculated by adding 100ng/L of standard to each sample. The samples were processed according to the method described above to ensure reliability of the experimental data. The recoveries of blank water samples with standard samples were 61.30–152.4% and RSD < 15% (Table S3). The target antibiotic matrix effect values were 88–119%. The minimum detection limit was measured at a signal-to-noise (S/N) ratio of 3 and the minimum quantification limit was measured at an S/N ratio of 10. Detailed information is provided in Table S3. Therefore, the method had good reproducibility, and its precision and accuracy meet the requirements for the analysis of antibiotics in samples.Therefore, in combined with previous studies, the correlation between antibiotic concentration and ARG was analyzed (Fig. 8)[51].

The results showed that the concentrations of macrolides (MLs) and sulfonamides(SAs) were significantly correlated with the corresponding ARGs. However, antibiotic concentrations can also show significant correlations with non-corresponding ARGs. For example, the concentrations of quinolone(QN) antibiotics in wastewater samples were significantly correlated with tetracycline(TC) ARGs and pnenicol ARGs. One reason may be that previous studies have only quantified and analyzed the concentrations of 19 antibiotics in four classes, ignoring the presence of but not tested antibiotics in wastewater and soil. Another reason is that the co-existence of heavy metal ions and various types of antibiotics may cause co-selection and cross-selection to microorganisms[52,53]. The significant correlation indicates that the presence of antibiotics is important to the selection of ARGs, and ARG is very easy to enrich when exposed to a certain concentration of antibiotics. In addition, ARGs were also negative correlated with antibiotic concentrations, for example, SAs showed a significant negative correlation with pnenicol ARGs. This negative correlation suggests that ARGs are not necessarily the result of antibiotic



(A) Wastewater samples

(B) Soil samples

Fig. 8. Correlation analysis between antibiotic concentrations and ARGs, (**A**) wastewater samples, (**B**) soil samples. (*: $p < 0.05$) (Horizontal axis (X - axis) and Vertical axis (Y - axis): Both axes list different classes of antibiotics (or related ARG types). Each cell at the intersection of a row and a column in the heatmap represents the correlation coefficient between the two corresponding antibiotic classes (or ARG types). Color bar: The color gradient (from blue to red) represents the range of correlation coefficients. Blue indicates a negative correlation, red indicates a positive correlation, and the intensity of the color reflects the strength of the correlation.).

selection pressure, and it is important to identify potential environmental factors and stressors for ARGs[54]. The correlation between antibiotic concentrations and ARGs has been contradictory in different studies, such as Chen et al.[35] finding a significant positive correlation between TCs concentrations and tetracycline ARGs, while Gao et al.[52] found the opposite. Similar conclusions were found in soil samples. Studies have shown an asymptotic relationship between antibiotic concentration and ARGs abundance as wastewater irrigation time increases, and ARGs abundance does not change significantly over several years after decades of irrigation[55]. In addition to antibiotic concentrations, different physicochemical properties and microbial composition of soils have also been shown to affect the distribution of ARGs[56].

Conclusion

This study systematically investigated ARGs prevalence and microbial communities in different WWTSs and irrigated-soils. The main conclusions can be drawn:

(1) Residual ARGs at environmentally relevant levels (sufficient to pose dissemination risks) still exist in the effluent, especially in winter, which has become a “hotspot” for spreading ARGs to the environment. The AAO and AB processes increase the relative abundance and diversity of ARGs.

(2) Multidrug, tetracyclines, macrolides and aminoglycoside resistance genes were the main ARGs types. Seasonal variations did not cause significant differences in the distribution of ARGs.

(3) Compared to influent water, RTP is more effective in reducing the relative abundance of ARGs rather than relying on biological treatment. Additionally, the efficiency of reducing the relative abundance of ARGs in summer is higher than that in winter.

(4) Due to frequent irrigation, soil samples from RTP had the highest relative abundance of ARGs. These results suggest that agricultural activities can enrich ARGs.

(5) The microbial genera significantly related to ARG may become its potential host, such as *Rhodanobacter* had significant correlations with *ropB2*, *carA* and *oleB* in wastewater samples.

Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are available from the corresponding author upon reasonable request. The datasets generated and analysed during the current study are available in the National Center for Biotechnology Information (NCBI) repository, persistent web link is http://www.ncbi.nlm.nih.gov/bioproject/824545, accession number to datasets is PRJNA824545.

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Author contributions

Hua Fang: ** Investigation, Supervision, Conceptualization, Writing - review & editing, Funding acquisition. **Miao Pu: ** Investigation, Methodology, Conducting the experiment, Data collation and analysis, Writing - review & editing. **Ailier jiang: ** Supervision, Writing - review & editing. **Fuer haiti: ** Supervision, Writing - review & editing. **Yubei Liu: ** Supervision, Writing - review & editing. **Nuerla Ailijiang: ** Investigation, Supervision, Conceptualization, Writing - review & editing, Formal analysis, Writing - review & editing, Funding acquisition. **Anwar Mamat: ** Investigation, Formal analysis. **Xifeng Tu: ** **Supervision, Writing - review & editing..

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to N.A.

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